

Day 12 Student Handout

Geometry Summer Credit | Special Right Triangles

Name:	Date:	Period:
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Learning Targets

I can use 45-45-90 and 30-60-90 side ratios. I can decide when a special right triangle pattern is faster than the Pythagorean theorem. I can write exact answers with radicals.

Lesson 12A: 45-45-90 Triangles

Warm-Up

1.	Simplify: $\sqrt{72}$.
2.	Find the hypotenuse of a right triangle with legs 6 and 8.
3.	Find the distance between (0, 0) and (5, 12).

Concept Launch

45-45-90 Pattern

A 45-45-90 triangle is an isosceles right triangle. The legs are equal. If each leg is x , the hypotenuse is $x\sqrt{2}$. Ratio: $x : x : x\sqrt{2}$.

Problem	Work / notes
A 45-45-90 triangle has legs 7 and 7. Find the hypotenuse.	Pattern: Answer:
A square has side length 10. Find the diagonal.	The diagonal creates: Answer:
A 45-45-90 triangle has hypotenuse $8\sqrt{2}$. Find each leg.	$x\sqrt{2} = 8\sqrt{2}$ Legs:

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Lesson 12B | 30-60-90 Triangles

Lesson 12B: 30-60-90 Triangles

30-60-90 Pattern

In a 30-60-90 triangle, the shortest side is across from 30 degrees. If the short leg is x , then the long leg is $x\sqrt{3}$, and the hypotenuse is $2x$.

Problem	Work / notes
A 30-60-90 triangle has short leg 5. Find the long leg and hypotenuse.	Short leg x : Long leg: Hypotenuse:
A 30-60-90 triangle has hypotenuse 18. Find both legs.	$2x = 18$ Short leg: Long leg:
A 30-60-90 triangle has long leg $6\sqrt{3}$. Find the short leg and hypotenuse.	$x\sqrt{3} = 6\sqrt{3}$ Short leg: Hypotenuse:
Which tool would you use: special pattern or Pythagorean theorem? A right triangle has acute angles 30 and 60.	Decision: Reason:

Practice

6.	A 45-45-90 triangle has leg 11. Find the hypotenuse.
7.	A 30-60-90 triangle has short leg 8. Find the long leg and hypotenuse.
8.	A 30-60-90 triangle has hypotenuse 22. Find the short leg and long leg.
9.	A 45-45-90 triangle has hypotenuse $14\sqrt{2}$. Find each leg.